HANA BERHE HAGOS

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PROFESSIONAL SUMMARY:

Aspiring Machine Learning Engineer and Cybersecurity Analyst with practical experience in AI/ML development, defensive/offensive security, and data analytics. Completed ALX Cybersecurity program and INSA CyberTalentCenter. Kifiya AI Mastery Programme, and University of Tokyo Data science certification. Experienced in building security tools, deploying ML models, and analyzing threats across cloud, endpoint, and network environments. Passionate about applying AI to cybersecurity challenges, particularly in the Ethiopian context.

TECHNICAL SKILLS

Machine Learning & AI: Python, scikit-learn, XGBoost, Random Forest, LangChain, Sentence Transformers, SHAP, SMOTE

Cybersecurity: Active Directory Security, BloodHound, MITRE ATT&CK, SIEM (Splunk, KQL), EDR, Incident Response, Threat Hunting, Malware Analysis, Vulnerability Management, Penetration Testing, OSINT, Nmap, Burp Suite, Metasploit, Wireshark

Data & Databases: SQL, Pandas, NumPy, Matplotlib, dbt, PostgreSQL, MongoDB, ChromaDB

Development & DevOps: Git, Docker, FastAPI, Gradio, MLflow, Dagster, PowerShell, Base

WORK EXPERIENCE

- ✓ **Cybersecurity Trainee | ALX Africa | Remote | 2026** Completed intensive 6-month cybersecurity program covering offensive security, defensive security, and strategic security
  - Achieved 94%+ average across all modules including network penetration testing, Active Directory exploitation, incident response, threat hunting, and cloud security
  - Built and deployed security tools and completed hands-on labs in SIEM, EDR, log analysis, and malware analysis
  - Earned 100% on all knowledge checks and capstone projects
- ✓ **Cybersecurity Summer Camp Participant | INSA (Information Network Security Administration) | Ethiopia | 2026** Selected for Ethiopia's national cybersecurity summer camp program
  - Received hands-on training in network security, threat detection, and incident response from national experts
  - Collaborated on team projects addressing Ethiopian cybersecurity challenge
  - Currently contributing to ongoing security projects and research initiative
- ✓ **Data & AI Trainee | 10 Academy, Kifiya AI Mastery Programme | Addis Ababa | 202**6 Reduced model training time by 30% by implementing automated feature engineering pipelines for 10+ machine learning projects
  - Enabled real-time predictions by deploying 5 machine learning models as REST APIs using FastAPI and Docker
  - Improved data processing efficiency by 40% by building reproducible data pipelines with dbt and Dagster
  - Maintained 100% on-time submission rate across 12 weekly assignments through effective use of Git branching and CI/CD workflows

PROJECTS

Credit Risk Model | *Python, scikit-learn, FastAPI*

Built scoring model using RFM analysis. Random Forest achieved 99% accuracy. Deployed as REST API.

github.com/Hannaberhe/credit-risk-model

Fraud Detection System | *Python, SHAP, SMOTE*

Detected fraud on imbalanced data (0.17% fraud rate). Achieved F1=0.68. Used SHAP for explainability.

github.com/Hannaberhe/fraud-detection

RAG Complaint Chatbot | *LangChain, ChromaDB, Gradio*

Built chatbot answering questions from complaint narratives using vector search and semantic retrieval.

github.com/Hannaberhe/rag-complaint-chatbot

EDUCATION : B.Sc. Software Engineering | Addis Ababa Science and Technology University | 2024-Present

Cybersecurity Program and Data Analysis | ALX Africa | 2026

INSA Cybersecurity Summer Camp | Information Network Security Administration | 2026

Kifiya AI Mastery Programme | 10 Academy | 2026

CERTIFECATES

AI Mastery — 10 Academy (Kifiya), 2026

Cybersecurity — UNDP and ALX Africa 2026

Data Analytics — ALX Africa, 2026

Data Analytics and Visualization — INFNOVA Technology, 2026

Data Science — GCI World, University of Tokyo, April 2026

REFERENCES

Goitom Mekonnen(PhD) : Founder and CEO of TigraiCodes +254704308694, goitomg2m@gmail.com